

MUHAMMAD ARRY NUGROHO

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Fresh graduate in Geophysical Engineering student at Bandung Institute of Technology (GPA: 3.38/4.00) with hands-on experience in seismic, gravity, magnetic, and resistivity methods. Proven ability to lead teams and deliver technical results through internship experience and multiple field projects.

Pendidikan

Bandung Institute of Technology - Bandung

Mar 2022 - Oct 2026 (Perkiraan)

Undergraduate Geophysical Engineering in Faculty of Mining and Petroleum Engineering, 3.38/4.00

- Undergraduate Thesis (2025–2026): "Pore Pressure Prediction and Overpressure Zone Characterization in the Poseidon Field, Browse Basin, Australia using Eaton and Kan & Swan Methods through Well Log and Seismic Data Integration." Successfully created a 3D pore pressure distribution map with overpressure characterization of the Jamieson-Plover Formation at the Poseidon Field, Australia, using the Eaton and Kan & Swan methods.

Work Experience

PT. Antam Tbk - Jakarta, Indonesia

Jun 2025 - Jul 2025

Internship

- Conducted drone-based magnetic data acquisition (dronemagnetic method) at Block X, Pacitan, East Java, operating a DJI Matrice 300/350 RTK integrated with a MagArrow II UAV-enabled magnetometer from Geometrics
- Executed end-to-end data processing pipeline including diurnal correction, IGRF correction, tie-line correction, gap adjustment, B-spline filtering, tie line levelling, and microleveling to produce clean Total Magnetic Intensity (TMI) maps

Organizational and Committee Experience

HIMA TG "TERRA" ITB - Bandung

Feb 2025 - Jan 2026

- Led the Interest and Talent Division, designing and executing development programs to enhance the technical and professional competencies of geophysical engineering students at ITB.

Society of Exploration Geophysicists Student Chapter ITB - Bandung

May 2024 - Dec 2024

- Managed the Field Trip Division, coordinating 3 annual industry exposure programs, SEG Field Camp, Company Visit, and Excursion
- Facilitated geophysics students' direct exposure to real-world professional environments through structured field activities involving 80 participants

Character Development Training ITB 2024 - Bandung

Sep 2024 - Feb 2025

- Served as trainer and mentor for 40 incoming students in ITB's Character Development Training
- Delivered sessions on adaptation, integrity, and humility to support new students' transition into the university environment

Indonesian Undergraduate Geophysics Competition (IUGC) - Bandung

Aug 2024 - Sep 2024

- Acted as Liaison Officer bridging communication between organizing committee and competition participants from 5 universities
- Ensured smooth coordination of participant preparations throughout the event

Other Experience and Skill

- Projects - Gravity and Magnetic at Karangsambung** (2025): Served as field operator for gravity and magnetic data acquisition in the Karangsambung area, covering jatibungkus region. Performed end-to-end data processing and contributed to geological interpretation of subsurface structures to identify intrusion zone.
- Community Service Project — Aquifer Mapping** (2025): Led a field team of 15 members in conducting resistivity surveys using a McOHM resistivity meter to map groundwater distribution in a rural village. Demonstrated decisive leadership under extreme field conditions by implementing real-time equipment safety protocols, ensuring both data quality and instrument integrity throughout the survey.
- Practicum Assistant - Seismology** (2025): Guided 31 students per semester through hands-on passive seismic analysis, covering P- and S-wave picking, hypocenter and magnitude determination, and fault projection using manual methods, geophysical software, and Python programming.
- Practicum Assistant - Geophysical Instrumentation 1** (2024): Guided 31 students in designing and building IoT-based geophysical instrumentation projects, integrating ESP32 microcontrollers with various sensors to produce functional field-ready devices.
- Project - Geophysical Acquisition & Instrumentation** (2024): Conducted seismic reflection surveys, operating a weight-drop seismic source across 5 survey lines, supporting full field logistics to ensure efficient and accurate data collection, and developed IoT-based geophysical instruments using ESP32, ADXL345, and INA219 for seismic and resistivity measurements.
- Hard Skills:** Familiar with using software Seisgram2k, IPI2Win, Res2Dinv, Stereonet, Surfer, Hampson Russel, Petrel, Paleoscan, etc